

MLOnEdge Phase_01 document

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Title	LogiEdge: Intelligent Edge AI Platform for Cold-Chain Logistics
Repository path	https://github.com/2024ac05306-droid/MLOnEdge_miniproject

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Project folder and purpose

Development environment (Git, Source tree, and VS code)

Project folder and purpose

The architecture contains following files and folders

Folder and file Description:

.github/	GitHub Actions workflows for CI/CD automation.
data/	Stores input data and generated preprocessing artifacts.
k8s/	Kubernetes manifests for deploying the model serving API.
logs/	Stores application, training, and API request logs.
models/	Stores trained model artifacts.
mqtt_architecture	Explained MQTT architecture in detail
data_pipeline	Simulation, data generation and normalization
simulator.py	Temperature, vibration and door events are simulated using local MQTT
preprocessing	Simulated data is preprocessed for the 3 anomalies
normalization	Preprocessed data is normalized with mean and standard deviation
outputs/	Stores generated reports, plots, predictions, and MLflow artifacts.
Tests/	Unit tests for preprocessing and model training code.
training/train_model.py	
Data pipeline/preprocessing.py	Handles data preprocessing.
inference/inference_service.py	Runs model inference/prediction workflows.

Readme.md	Project information
Project structure.md	Folder structure of the project
Config.py	Helper functions defined
utils.py	Helper functions defined
Dockerfile	Container image definition for the API application.
requirements.txt	Python package dependencies.
Makefile	Common project commands for install, test, train, serve, Docker, and Kubernetes.

Pre-process

```

ics due to floating-point round-off errors from different computation orders. To turn them off, set the
_OPTS=0`.
MLflow URI: sqlite:///mlflow.db
MLflow Experiment: G44_logibridge_miniproject

Extracting features across all operational modes...
Saved data/normal_features.csv | Shape: (108, 6)
Saved data/warning_features.csv | Shape: (109, 6)
Saved data/critical_features.csv | Shape: (129, 6)

Computing Normal-class training stats...
[INFO] Saved clean Normal training statistics to: data\training_stats.npy
[INFO] Saved clean Normal training statistics to: outputs\training_stats.npy
[INFO] Saved clean Normal training statistics to: E:\MTech\Sem_3\MLOnEdge\training_stats.npy

Feature extraction and stats generation completed successfully.
PS E:\MTech\Sem_3\MLOnEdge>

```

Figure 1: Pre-processed data

```

PS E:\MTech\Sem_3\MLOnEdge> C:/Users/hp/miniconda3/envs/logibridge_py311/python.exe e:/MTech/Sem_3/MLOnEdge/data_pipeline/normalization.py
2026/08/09 00:16:08 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2026/08/09 00:16:09 INFO mlflow.store.db.utils: Updating database tables
INFO [alembic.runtime.migration] Context impl SQLiteImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.
INFO [alembic.runtime.migration] Context impl SQLiteImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.

Training statistics saved:
data\training_stats.npy

Mean:
[ 3.83127419  0.44690707 -0.04931851  0.41798987  0.47480556  4.14581082]

Std:
[0.05057727 0.02722163 0.28842041 0.0116926  0.01787451 0.44450898]

```

Figure 2: Normalization

```
60 DATA DTR - BASE DTR / "data"
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Std:
[0.05057727 0.02722163 0.28842041 0.0116926 0.01787451 0.44450898]
PS E:\MTech\Sem_3\MLOnEdge> C:\Users\hp\miniconda3\envs\logibridge_py311\python.exe e:\MTech\Sem_3\MLOnEdge\training\generate_dataset.py
2026/08/09 00:19:35 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2026/08/09 00:19:35 INFO mlflow.store.db.utils: Updating database tables
INFO [alembic.runtime.migration] Context impl SQLiteImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.
INFO [alembic.runtime.migration] Context impl SQLiteImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.

Dataset generated:
E:\MTech\Sem_3\MLOnEdge\data\training_dataset.csv

Class distribution:
label
0 108
1 109
2 129
Name: count, dtype: int64

MLflow logging completed.
PS E:\MTech\Sem_3\MLOnEdge>
```

Figure 3: data generation

```
Loading training dataset...
Dataset Shape : (402, 7)
Loading normalization statistics...
Training Samples : 321
Validation Samples : 81
Loading trained MLP model...
WARNING:tensorflow:From C:\Users\hp\miniconda3\envs\logibridge_py311\Lib\site-packages\tf_keras\src\l
erly_outside_functions is deprecated. Please use tf.compat.v1.executing_eagerly_outside_functions in

Model: "sequential"

-----
Layer (type)                Output Shape                Param #
-----
dense (Dense)                (None, 32)                  224
dense_1 (Dense)              (None, 16)                  528
dense_2 (Dense)              (None, 3)                   51
-----

Total params: 803 (3.14 KB)
Trainable params: 803 (3.14 KB)
Non-trainable params: 0 (0.00 Byte)
```

```
Epoch 9/100
+1m21/21+0m +32m-----+0m+37m+0m +1m0s+0m 10ms/step - accuracy: 0.9844 - loss: 0.0788 - val_accuracy: 0.9753 - val_loss: 0.3781
Epoch 10/100
+1m21/21+0m +32m-----+0m+37m+0m +1m0s+0m 10ms/step - accuracy: 0.9844 - loss: 0.0652 - val_accuracy: 0.9753 - val_loss: 0.3948
Epoch 11/100
+1m21/21+0m +32m-----+0m+37m+0m +1m0s+0m 10ms/step - accuracy: 0.9813 - loss: 0.0623 - val_accuracy: 0.9753 - val_loss: 0.3994
Epoch 12/100
+1m21/21+0m +32m-----+0m+37m+0m +1m0s+0m 10ms/step - accuracy: 0.9844 - loss: 0.0700 - val_accuracy: 0.9630 - val_loss: 0.4176
Epoch 13/100
+1m21/21+0m +32m-----+0m+37m+0m +1m0s+0m 10ms/step - accuracy: 0.9844 - loss: 0.0570 - val_accuracy: 0.9753 - val_loss: 0.4364

Validation Accuracy : 95.06%

Classification Report

      precision    recall  f1-score   support

   Normal         0.93       0.96       0.95         27
   Warning         1.00       0.93       0.96         27
   Critical         0.93       0.96       0.95         27

 accuracy          0.95
 macro avg          0.95
weighted avg          0.95

Model saved : models\best_model.keras
2026/07/30 02:15:16 WARNING mlflow.models.model: `artifact_path` is deprecated. Please use `name` instead.
2026/07/30 02:15:16 WARNING mlflow.tensorflow: You are saving a TensorFlow Core model or Keras model without a signature. Inference with mlflow.pyfunc.spark_udf() will
not work unless the model's pyfunc representation accepts pandas DataFrames as inference inputs.
2026/07/30 02:16:05 WARNING mlflow.models.model: Model logged without a signature and input example. Please set `input_example` parameter when logging the model to auto
infer the model signature.

Training completed successfully.
Validation Accuracy : 95.06%

(logibridge) E:\MTech\Sem_3\MLOnEdge>
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Removing pruning wrappers...
Saved : E:\MTech\Sem_3\MLOnEdge\models\best_model_pruned.keras
Model: "sequential"

-----
Layer (type)                Output Shape              Param #
-----
dense (Dense)                (None, 32)                224
dense_1 (Dense)              (None, 16)                528
dense_2 (Dense)              (None, 3)                 51
-----

Total params: 803 (3.14 KB)
Trainable params: 803 (3.14 KB)
Non-trainable params: 0 (0.00 Byte)
-----

Model Size
-----
Original : 0.03 MB
Pruned   : 0.01 MB

Model logged to MLflow.

=====
Model Pruning Completed Successfully
=====

Saved Model : E:\MTech\Sem_3\MLOnEdge\models\best_model_pruned.keras
```

